ROBUST DESIGN AND PRODUCT SPECIFICATION COURSE PLAN

To transform student projects into professional technical documentation and digital dossiers

Session	Title	Objective	Content	Activities	Deliverable / Evidence
1	Robust Design Diagnosis	Identify the current technical status of each project and detect risks, gaps and missing information.	Robust design principles, concept vs. prototype vs. MVP, technical diagnosis, risk identification, project documentation workflow.	Teams collect previous project materials, upload them to their project notebook, and create a first technical diagnosis.	Robust design matrix + technical gap analysis
2	Design Requirements and Specifications	Convert research, user needs and feedback into clear product requirements.	User requirements, market requirements, functional requirements, technical requirements, ergonomic requirements, production and sustainability criteria.	Teams organize their information into a requirements table and define criteria for success.	Product specification / requirements table
3	Materials and CMF	Justify material, color and finish decisions according to function, production and user experience.	Material selection, properties, durability, cost, manufacturing compatibility, CMF palette, finishes and textures.	Teams create material sheets and compare material/process alternatives for their main components.	CMF palette + material selection table
4	Technical Drawings and BOM	Organize the product as a system of parts, assemblies, components and production information.	Technical drawings, part drawings, assembly drawings, exploded views, dimensions, tolerances, components, BOM and OpenBOM.	Teams define all parts, separate manufactured and commercial components, and start their BOM.	Preliminary drawings + BOM / OpenBOM
5	Ergonomics, Use Diagrams and MVP	Translate ergonomic analysis and user interaction into visual documentation and validation criteria.	Body-product relationship, posture, grip, reach, points of contact, use sequence, user errors, comfort, safety, MVP definition.	Teams create use diagrams, ergonomic diagrams and define what their MVP includes and excludes.	Use diagrams + ergonomic analysis + MVP definition
6	Production Manual and Sustainability	Explain how the product can be manufactured, assembled, maintained and improved.	Production plan, assembly sequence, tools, processes, quality control, maintenance, repairability, sustainability, life cycle and basic regulatory references.	Teams organize the production flow from materials to manufacturing, assembly, inspection, use, maintenance and end of life.	Production manual + life cycle / sustainability plan
7	Final Integration and Digital Dossier	Compile all documentation into a final technical product specification package.	Final report structure, visual hierarchy, website organization, AI-assisted review, Notion, Google Sites, Gamma, Canva, final presentation.	Teams review, edit, complete and publish their final report as a website or digital dossier.	Final technical report + website / digital dossier